Area & Differential equation (4)

Q1. Let K be a positive integer and $f(x)$ be a polynomial with integer coefficients satisfying $2\int_1^x f(t)dt + x^K = x f(x)$, where $x \geq 1$. Find the sum of possible values of K.

P: If $y = f_1(x)$ and $y = f_2(x)$ are two solutions of the equation $y\,dx + dy = -e^x y^2 dy$ where $f_1(0) = 1$ & $f_2(0) = -1$.

Q2. The number of solution(s) of the equation $f_1(x) f_2(x) + x^2 = 0$ is (are)

(A) 0 (B) 1 (C) 2 (D) 3

Q3. If area enclosed by $y = f_1(x)$, $y = e^x$ and the x-axis is A_1 and $y = f_2(x)$, $y = -e^x$ and the x-axis is A_2 then $A_1 + A_2$ is equal to

(A) $6\ln 2$ (B) 3 (C) 6 (D) $3\ln 2$.

Q4. Let $\frac{dy}{dx} + y = f(x)$ where y is a continuous function of x with $y(0) = 1$ and $f(x) = \begin{cases} e^{-x}, & 0 \leq x \leq 2 \\ \alpha e^{-2} & x > 2 \end{cases}$

Which of the following hold(s) good?

(A) $y(1) = 2e^{-1}$ (B) $y'(1) = e^{-1}$ (C) $y(3) = -2e^{-3}$ (D) $y'(3) = -2e^{-3}$.

Q5. Let function $y = f(x)$ satisfies the differential equation $x^2 \frac{dy}{dx} = y^2 e^{\frac{1}{x}}$ $(x \neq 0)$ and $\lim_{x \to 0} f(x) = 1$

Identify the correct statement(s)

(A) Range of $f(x)$ is $(0,1) - \{\frac{1}{2}\}$

(B) $f(x)$ is bounded (C) $\lim_{x \to \pm\infty} f(x) = 1$ (D) $\int_0^e f(x)\,dx > \int_0^1 f(x)\,dx$

~~Q6. Let $y = f(x)$ defined on R satisfies~~

P: Let C be a curve $f(x,y) = 0$ passing through $(1,1)$ such that it is orthogonal to family of circles $x^2 + y^2 - ax = 0$ where 'a' being parameter.

Q6. If the curve C intersects the circle $x^2 + y^2 + 2x + 4y + K = 0$ orthogonally then K is equal to

(A) 4 (B) 2 (C) 0 (D) -4

Q7. The co-ordinates of a point $P(x,y)$ are functions of time 't' the locus of point $P(x,y)$ is a curve given by $\frac{dx}{dt} + \frac{dy}{dt} = t$ and $\frac{dx}{dt} - 2\frac{dy}{dt} = t^2$ at any instant of time t, the locus of point $P(x,y)$ is a curve given by (assume $x(0)=0, y(0)=0$)

(A) $(x+y)^3 = (x-2y)^3$ (B) $x = \frac{t^2}{6} - \frac{t^3}{9}, y = \frac{t^2}{3} + \frac{t^3}{9}$

(C) $8(x+y)^3 = 9(x-2y)^2$ (D) $9(x+y)^3 = 8(x-2y)^2$.

Q8. The solution of the differential equation $\frac{\sqrt{x}\,dx + \sqrt{y}\,dy}{\sqrt{x}\,dx - \sqrt{y}\,dy} = \sqrt{\frac{y^3}{x^3}}$ is given by

(A) $\frac{3}{2}\log\left(\frac{y}{x}\right) + \log\left(\frac{x^{3/2}+y^{3/2}}{x^{3/2}}\right) + \left(\tan^{-1}\left(\frac{y}{x}\right)\right)^{3/2} + c = 0$

(B) $\frac{2}{3}\log\left(\frac{y}{x}\right) + \log\left(\frac{x^{3/2}+y^{3/2}}{x^{3/2}}\right) + \tan^{-1}\left(\frac{y}{x}\right) + c = 0$

(C) $\frac{2}{3}\left(\frac{y}{x}\right) + \log\left(\frac{x+y}{x}\right) + \tan^{-1}\left(\frac{y^{3/2}}{x^{3/2}}\right) + c = 0$

(D) $\frac{1}{2}\log(x^3+y^3) + \tan^{-1}\left(\frac{y}{x}\right)^{3/2} = \frac{c}{2}$.

Q9. Let $\triangle ABC$ be a triangle with vertices $A(6, 2\sqrt{3}+2)$, $B(4,2)$ and $C(8,2)$. Let R be the region consisting of all those points P inside triangle $\triangle ABC$ which satisfy $d(P,BC) \geq \max\{d(P,AB), d(A,AC)\}$ where $d(P,L)$ denotes the distance of the point P from the line L. Then the area of the region R is (in square units)

(A) $\frac{2\sqrt{3}}{3}$ (B) $\frac{4\sqrt{3}}{3}$ (C) $2\sqrt{3}$ (D) $4\sqrt{3}$.

Q10. The solutions of the differential equation $\frac{dy}{dx} = \frac{1}{xy(x^2\sin y^2 + 1)}$ is

(A) $x^2(\cos y^2 - \sin y^2 - 2ce^{-y^2}) = 2$

(B) $x^2(\cos y^2 - \sin y^2 - 2e^{-y^2}) = 4c$

(C) $y^2(\cos x^2 - \sin x^2 - 2ce^{-y^2}) = 2$

(D) $y^2(\cos x^2 - \sin x^2 - ce^{-y^2}) = 4c$.

Q11. Suppose a solution of the differential equation $(xy^3 + x^2y^7)\frac{dy}{dx} = 1$ satisfies the initial condition $y(\frac{1}{4}) = 1$ then the value of $\frac{dy}{dx}$ when $y = -1$ is —

(A) $-\frac{3}{20}$ (B) $-\frac{20}{3}$ (C) $-\frac{5}{16}$ (D) $-\frac{16}{5}$.

Q12. A function $y = f(x)$ satisfies the differential equation $\frac{dy}{dx} - y = \cos x - \sin x$ with initial condition that y is bounded when $x \to \infty$, then the area enclosed by $y = f(x)$, $y = \cos x$ and the y-axis in first quadrant is

(A) $\sqrt{2} - 1$ (B) $\sqrt{2}$ (C) 1 (D) $\frac{1}{\sqrt{2}}$

Q13. Let $f(x)$ be a non-negative continuous function such that the area bounded by the curve $y = f(x)$, x-axis and the ordinates $x = \frac{\pi}{4}$, $x = \beta > \frac{\pi}{4}$ is $\beta \sin\beta + \frac{\pi}{4}\cos\beta + \sqrt{2}\beta$. then $f'(\frac{\pi}{2}) =$

(A) $\frac{\pi}{2} - \sqrt{2} - 1$ (B) $\frac{\pi}{2} + \sqrt{2} - 1$ (C) $-\frac{\pi}{2}$ (D) $1 - \frac{\pi}{4} + \sqrt{2}$.

Q14. Let $f : (0, \infty) \to (0, \infty)$ be a differentiable function satisfying

$x\int_0^x (1-t) f(t)\,dt = \int_0^x t f(t)\,dt$, $\forall x \in R$ & $f(1) = 1$ then

(A) $f(2) = \frac{1}{8}e^{1/2}$ (B) $f(2) = \frac{1}{8}e^{-1/2}$

(C) $f(3) = \frac{1}{27}e^{2/3}$ (D) $f(3) = \frac{1}{27}e^{-2/3}$.

Q15. The normal at a general point (a, b) on curve makes an angle θ with positive direction of x-axis, which satisfies $b(a\tan\theta - \cot\theta) = a(b^2 - 1)$ then equation of the curve can be

(A) $y = e^{x^2/2} + c$ (B) $\log_e(|y|) = x^2$

(C) $y = ke^{x^2/2}$ (D) $x^2 + y^2 = k$.

Q16. Let $y = f(x)$ be a polynomial such that

$$\int_{x+\sqrt{5}}^{x+\sqrt{2}} f(x)\,dx = ax + b \quad \forall\, x \in R \quad (a \neq 0,\ b \in R)$$

and $f(0) = 3$, $f(1) = 5$ Then which of the following is/are true?

(A) $y = f(x)$, $f: R \to R$ is one-one & onto

(B) $y = f(x)$, $f: R \to R$ is neither one-one nor onto

(C) Area bounded by the curve $y = f(x)$ between the lines $x = 0$, $x = 2$, x-axis is 10 square units

(D) Area bounded by the curve $y = f(x)$ between the lines $x = 0$, $x = 2$, x-axis is 8 square units.

Q17. Given a function g which has a derivative $g'(x)$ for every real x and which satisfies the following conditions $g'(0) = 2$ & $g(x+y) = e^x g(y) + e^y g(x)$ for all x & y, then which of the following is/are correct

(A) $g(2x) = 2e^x g(x)$ & $g(3x) = 3e^{2x} g(x)$

(B) $g(nx) = ne^{(n-1)x} g(x)$ (C) $g(0) = 0$ & $\lim_{h \to 0} \frac{g(h)}{h} = 2$

(D) there is a constant c such that $g'(x) = g(x) + ce^x$ for all x.

P(18-19): $f(x)$ is a polynomial function such that $f(x^2+1) = f(x^2) + 2x^2$ and $f(0) = 0$

$g(x) = x f(x)$ Then

Q18. If $\int_0^1 g(x)\,dx = A$ then

(A) $A = 2\int_0^1 f(x)\,dx \neq 0$ (B) $|A| = 1$

(C) $A + |A| = 0$ (D) $\frac{1}{2} < |A| < 1$

Q19. Area bounded by $\min\{f(x), g(x)\}$, $x = 2$ & x-axis in first quadrant is

(A) 11/2 (B) 5/6 (C) 11/2 (D) 4/3.

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Q 20. Slope of the tangent at any point (x,y) on the curve $y = f(x)$ is $\frac{y}{x^2}$ and the curve passes through the point $(1,3)$ then $\lim_{x\to\infty} f(x) = ke$ where $k =$

Q 21. Let A and B be the points of intersection of the parabola $y = x^2$ and the line $y = x+2$ and let C be the point on the parabola where the tangent line is parallel to the graph of $y = x+2$ then

$$\frac{(\text{area of the parabolic segment cut from the parabola by the line})}{\text{area of the triangle}} = \frac{m}{n}$$

then $m+n =$

Q 22. If $p(x) = x^3 - 3b^2x + 16$ if $p(x) = 0$ has all integral roots (no complex root) and local minimum value of $p(x)$ is 0. And the area bounded by the curve $y = p(x)$ & x-axis is then $\left[\frac{\sqrt{k}}{2}\right]$ . . . (where [.] is G.I.F.)

Q 23. Let $F(x) = \int_x^{x^2+\pi/6} 2\cos^2 t\, dt$ $\forall x \in R$ and $f: [0, \frac{1}{2}] \to [0, \infty)$ be a continuous function. For $a \in [0, \frac{1}{2}]$, if $F'(a) + 2$ is the area of the region bounded by $x = 0$, $y = 0$, $y = f(x)$ and $x = a$ then $f(0)$ is

Q 24. If c is a constant such that the line joining the points $(0,3)$, $(5,-2)$ is a tangent to the curve $y = \frac{c}{x+1}$ then the area of the region bounded by the curve & x-axis and $x = 1$, $x = 2$ is $\alpha\{1 - \frac{1}{2.2} + \frac{1}{2^2.3} - \frac{1}{2^3.4} + \dots \infty\}$ where α equals to

Q25. Let $S_1 = \{(x,y) : \log_{4-x}(16x-y^2)$ is defined$\}$

$S_2 = \{(x,y) : \frac{xy(x^2-3x+2)}{x^2-7x+12} > 0\}$

then area bounded by $S_1 \cap S_2$ is

(A) $\frac{8}{3}$ (B) $\frac{64}{3}$ (C) $\frac{32}{3}$ (D) none.

Q26. A square ABCD is inscribed in a circle of radius 4. A point P moves inside the circle such that $d(P, AB) \le \min\{d(P, BC), d(P, DA)\}$ where $d(P, AB)$ is the distance of a point P from the line AB. The area of region covered by the moving point P is

(A) 4π (B) 8π (C) $8\pi - 16$ (D) $4\pi - 4$

Q27. If $y_1(x)$ (not identically zero) is a solution of the differential equation $\frac{dy}{dx} + f(x)y = 0$ then a solution of the differential equation $\frac{dy}{dx} + f(x)y = r(x)$ is

(A) $\frac{1}{y_1(x)} \int y_1(x)\,dx$ (B) $y_1(x) \int \frac{r(x)}{y_1(x)}\,dx$

(C) $\int r(x)\, y_1(x)\,dx$ (D) none.

Q28. The primitive of the ~~region~~ differential equation $(2x\,y^4 e^y + 2xy^3 + y)\,dx + (x^2 y^4 e^y - x^2y^2 - 3x)\,dy = 0$ is

(A) $x^2 e^y + \frac{x^2}{y} - \frac{x}{y^3} = k$ (B) $x^2 e^y - \frac{x^2}{y} + \frac{x}{y^3} = k$

(C) $x^2 e^y + \frac{x^2}{y} + \frac{x}{y^3} = k$ (D) $x^2 e^y - \frac{x^2}{y} - \frac{x}{y^3} = k$

Q29. The area of the region $\{(x,y) : x^2+y^2 \le 5,\ ||x| - |y|| \ge 1\}$ is

(A) $4\{\pi - \tan^{-1}(\frac{24}{7})\} - 4$ (B) $10\pi - 4 - 20\sin^{-1}(\frac{2}{\sqrt{5}})$

(C) $10\pi + 4 - 20\sin^{-1}(\frac{2}{\sqrt{5}})$ (D) $2\{\pi - \tan^{-1}(\frac{24}{7})\} - 1$

Q30. Let $y(x)$ be a solution of the differential equation $(1+e^x)y' + ye^x = 1$. If $y(0) = 2$, then which of the following statements is/are true

(A) $y(4) = 0$ (B) $y(-2) = 0$ (C) $y(x)$ has a critical point

in the interval $(-1,0)$

(A) $y(x)$ has no critical point in the interval $(-1,0)$.

Q 31. Tangent drawn at any point P on a curve, meets x-axis at Q such that x coordinate of the circumcentre of triangle POQ is half of the y-coordinate of the circumcentre of the triangle POQ (where O is origin). Then the differential equation to such a curve is

(A) $\frac{dy}{dx} = \frac{x+2y}{2x-y}$ (B) $\frac{dy}{dx} = \frac{2y-x}{2x+y}$ (C) $\frac{dy}{dx} = \frac{2x+y}{2x-y}$

(D) None.

Q 32. If $P(x,y)$ be any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and S be the sectorial area bounded by the curve, x-axis and the line joining the origin to P, then

(A) $x = a\cos\left(\frac{2S}{ab}\right)$ (B) $y = b\sin\left(\frac{S}{ab}\right)$

(C) $S = \frac{ab}{2}\cos^{-1}\left(\frac{x}{a}\right)$ (D) $S = ab\sin^{-1}\left(\frac{y}{b}\right)$.

Q 33. Which of the following is/are true?

(A) If $y = f(x)$ is a strictly monotone function in (a,b) with $f'(x) \neq 0$, then the area bounded by $y = f(x)$, the lines $x = a$, $x = b$ & $y = f(c)$ where $c \in (a,b)$ is minimum when $c = \frac{a+b}{2}$.

(B) If $y = f(x)$ is strictly monotonic function in (a,b) with $f'(x) \neq 0$, then the area bounded by the lines $x = a$, $x = b$ & $y = f(c)$ where $c \in (a,b)$ is minimum when $c = \frac{2a+b}{3}$ or $c = \frac{a+2b}{3}$.

(C) If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ & the line $x = 0$, $x = 2$ & the x-axis is minimum when $a = 2/3$

(D) If the area bounded by $f(x) = \frac{x^3}{3} - x^2 + a$ & the line $x = 0$, $x = 2$ & the x-axis is minimum when $a = \frac{28}{81}$.

Q34. Consider the differential equation
$(x-2)(x-3)\frac{dy}{dx} + 2y = (x-1)(x-2)$

(A) All solutions do not tend to a finite limit as $x \to 2$

(B) All solutions tend to a finite limit as $x \to 2$

(C) All solutions tend to a finite limit as $x \to 3$

(D) No solution tends to a finite limit as $x \to 3$.

Q35. Consider the curve $y^2 = |x|$ & $x^2 = |y|$ then

(A) equation of the common tangent drawn to $y^2 = x$ & $x^2 = y$ is $x + y + \frac{1}{4} = 0$

(B) equation of the common tangent drawn to $y^2 = -x$, $x^2 = -y$ is $x + y - \frac{1}{4} = 0$

(C) area of the quadrilateral, formed by the common tangents (other than coordinate axes) to the curves $y^2 = |x|$ & $x^2 = |y|$ taken in pairs is $\frac{1}{8}$

(D) area bounded by the lines $|x| + |y| = \frac{1}{4}$ is $\frac{1}{8}$.

Q36. If the solution of the d.e
$(y + x\sqrt{xy}(x+y))\,dx + (y\sqrt{xy}(x+y) - x)\,dy = 0$
& $y(1) = 1$ is $\frac{x^2+y^2}{n} + m\left(\tan^{-1}\sqrt{\frac{x}{y}}\right) = 1 + \frac{\pi}{8}$
then the value $m + n =$ ___

Q37. If $y = f(x)$ is the solution of the differential equation $x(y^3 - x)\,dy = y(x + y^3)\,dx$ & $f(1) = (-2)^{1/3}$ & $f'(-2) = k$. Then $|k| =$ ___

Q38. If the solution of equation $\frac{dy}{dx} = y + \int_0^2 y\,dx$ is $y(x)$, given that $y(0) = 1$. Then the value of $[|y(2)|] =$ ___

Q38. A continuous function $f: R \to R$, satisfying the equation $f(x) = (1 + x^2)\left(1 + \int_0^x \frac{f^2(t)}{1+t^2}\,dt\right)$. If the area of triangle

formed by tangent drawn to the curve $y = f(x)$ at $x = 1$ with the coordinate axes is Δ then the value of $\left[\frac{\Delta}{3}\right]$ = (where [.] G.I.F.)

Q39.

Column I	Column II
(A) Area bounded by $f(x) = \cot^{-1}(\cos x)$ $0 \le x \le 2\pi$ with the tangent drawn to the curve $f(x) = \lvert\cos x\rvert$ at $x = \pi$	(P) $\frac{\pi^2}{2}$
(B) $y = f(x)$ be a function such that $f(x) = \min\{\sqrt{x(2-x)}, 2-x\}$, then area bounded by $y = f(x)$ & x-axis	(Q) $(\pi-1)^2$
(C) Area bounded by the curve $\lvert y\rvert = \sin^{-1}[\sin x]$, $2\pi < x < 3\pi$	(R) $\frac{\pi^2}{4}$
(D) Area bounded between curves $y = \tan^{-1}(\tan x)$, $\pi < x < 2\pi$ & the line $y = \pi - x$, $y = -x + 2\pi$	(S) $\frac{\pi}{4} + \frac{1}{2}$.

Q40. The solution of the differential equation

$x\,dy = (x^5 + x^3y^2 + y)\,dx$

1) $\tan^{-1}\left(\frac{y}{x}\right) = \frac{x^4}{4} + C$

2) $\tan^{-1}\left(\frac{y}{x}\right) = x^2 + C$

3) $\cot^{-1}\left(\frac{y}{x}\right) = \frac{y^4}{4} + C$

4) $\cot^{-1}\left(\frac{y}{x}\right) = \frac{x^4}{4} + x^2 + C$.

Q41. $y(x^2y + ex)\,dx - ex\,dy = 0$ then solution of above differential equation is

1) $x^3y + xy^3 = e^x + C$

2) $x^3y + 3e^x = 3cy$

3) $x^3y - 3e^x = C$

4) $xy^3 + xy^3 = 3cy$.

Q42. solution of the differential equation

$$\frac{dy}{dx} = \frac{3x^2y^4 + 2xy}{x^2 - 2x^3y^3}$$

1) $x^3y^2 + \frac{x^2}{y} = C$

2) $x^3y^2 + x^2 = y^2 + C$

3) $x^3y^2 + \frac{y^2}{x} = C$

4) $x^2y^3 + \frac{x^2}{y} = x^2y^2$.

Q 43. $\frac{dy}{dx} - y\frac{\phi'(x)}{\phi(x)} = \frac{-y^2}{\phi(x)}$ then $\phi(x) =$

1) $xy + cy$ 2) $xy - cy^2$ 3) $x^2y^2 + cy$ 4) $xy^3 + x^3y = cy$

Q 44. If a circle intersect with a concentric square at 8 - distinct points so that area of square outside the circle is equal to the area of the circle outside the square then ratio of radius of the circle to the side of the square

1) $\frac{1}{\pi}$ 2) $\frac{1}{\sqrt{\pi}}$ 3) $\frac{1}{\sqrt[3]{\pi}}$ 4) $\frac{1}{2\pi}$.

Answers (Assignment 4)

Q1 – 8

Q2 – B

Q3 – C

Q4 – A, D

Q5 – D A B D

Q6 – D

Q7 – C

Q8 – D

Q9 – B

Q10 – A

Q11 – D

Q12 – A

Q13 – C

Q14 – A, C

Q15 – B, C, D

Q16 – A C

Q17 – A B C D

Q18 – C

19 – B

Q20 – 3

Q21 – 7

Q22 – 5

Q23 – 3

Q24 – 2

Q25 – B

Q26 – A

Q27 – B

Q28 – C

Q29 – B

Q30 – C

Q31 – A

Q32 – A C

Q33 – A C

Q34 – B D

Q35 – A B C D

Q36 – 4

Q37 – 4

Q38 – 4

Q38 – 7

Q39 – A – Q, B – S, C – P, D – P

Q40 – A

Q41 – B

Q42 – A

Q43 – A

Q44 – B